

RELATÓRIO DE DETECÇÃO DE INCIDENTES - SOC PREDITIVO

Análise de ML, validação de modelos e mapeamento MITRE ATT&CK;

1. Resumo Executivo

Este relatório documenta 225 transações sinalizadas por detecção de anomalias (Elliptic Envelope, contamination=0.15). O classificador de triagem atingiu ROC-AUC de 0.99 em dados de teste. O modelo de severidade (regressão) obteve R²=0.66 e erro médio absoluto de 13.3 pontos em uma escala de 0-100.

2. Detalhes Técnicos dos Alertas de Alta Severidade

Ref/ID	Cliente (LGPD)	Valor	Horário	Score Predito	Prob. Suspeita
14560000	)-0000-4000-8000-0000C00000	R\$58,055.64	10:00h	92.8/100	100%
13750000	)-0000-4000-8000-0000C00000	R\$62,710.61	6:00h	62.5/100	100%
14030000	)-0000-4000-8000-0000C00000	R\$56,337.96	23:00h	42.7/100	100%
11500000	)-0000-4000-8000-0000C00000	R\$78,838.84	22:00h	85.9/100	100%
11060000	)-0000-4000-8000-0000C00000	R\$76,322.66	0:00h	100.0/100	100%
13520000	)-0000-4000-8000-0000C00000	R\$67,231.47	20:00h	66.7/100	100%
11060000	)-0000-4000-8000-0000C00000	R\$119,596.13	16:00h	100.0/100	100%
13500000	)-0000-4000-8000-0000C00000	R\$58,518.75	1:00h	77.8/100	100%
10990000	)-0000-4000-8000-0000C00000	R\$111,285.59	23:00h	82.5/100	100%
12720000	)-0000-4000-8000-0000C00000	R\$53,244.07	4:00h	71.3/100	100%
10570000	)-0000-4000-8000-0000C00000	R\$98,827.13	19:00h	77.1/100	100%
12040000	)-0000-4000-8000-0000C00000	R\$69,591.47	21:00h	80.3/100	100%
13030000	)-0000-4000-8000-0000C00000	R\$92,861.24	1:00h	89.0/100	100%
11300000	)-0000-4000-8000-0000C00000	R\$99,331.13	17:00h	92.4/100	100%
13300000	)-0000-4000-8000-0000C00000	R\$74,014.63	23:00h	81.8/100	100%
10030000	)-0000-4000-8000-0000C00000	R\$54,135.08	1:00h	74.6/100	100%
14700000	)-0000-4000-8000-0000C00000	R\$84,864.69	22:00h	64.6/100	100%
13050000	)-0000-4000-8000-0000C00000	R\$107,687.33	3:00h	100.0/100	100%
12030000	)-0000-4000-8000-0000C00000	R\$59,422.59	1:00h	65.6/100	100%
14340000	)-0000-4000-8000-0000C00000	R\$110,032.11	17:00h	100.0/100	100%
10070000	)-0000-4000-8000-0000C00000	R\$54,034.78	6:00h	83.6/100	100%
10030000	)-0000-4000-8000-0000C00000	R\$82,566.05	9:00h	75.2/100	100%
13270000	)-0000-4000-8000-0000C00000	R\$62,891.32	13:00h	83.4/100	100%
14030000	)-0000-4000-8000-0000C00000	R\$97,910.52	7:00h	100.0/100	100%
11700000	)-0000-4000-8000-0000C00000	R\$71,000.22	3:00h	72.5/100	100%

12350000	)-0000-4000-8000-0000C0000R01	PS59,008.53	6:00h	74.1/100	100%
13040000	)-0000-4000-8000-0000C0000R01	PS92,124.24	8:00h	82.9/100	100%
11030000	)-0000-4000-8000-0000C0000R01	PS75,278.75	3:00h	76.1/100	100%
14000000	)-0000-4000-8000-0000C0000R01	PS100,177.76	20:00h	91.6/100	100%
12970000	)-0000-4000-8000-0000C0000R01	PS81,241.57	5:00h	78.7/100	100%
10270000	)-0000-4000-8000-0000C0000R01	PS58,704.64	7:00h	60.8/100	100%
14550000	)-0000-4000-8000-0000C0000R01	PS84,987.58	23:00h	65.0/100	100%
12000000	)-0000-4000-8000-0000C0000R01	PS77,513.57	16:00h	66.0/100	100%
14930000	)-0000-4000-8000-0000C0000R01	PS60,702.95	5:00h	64.0/100	100%
12300000	)-0000-4000-8000-0000C0000R01	PS90,142.81	1:00h	88.8/100	100%
12340000	)-0000-4000-8000-0000C0000R01	PS119,358.79	5:00h	100.0/100	100%
10540000	)-0000-4000-8000-0000C0000R01	PS78,189.68	23:00h	73.3/100	100%
13570000	)-0000-4000-8000-0000C0000R01	PS114,444.33	7:00h	100.0/100	100%
10750000	)-0000-4000-8000-0000C0000R01	PS69,030.85	16:00h	60.3/100	100%
12430000	)-0000-4000-8000-0000C0000R01	PS115,804.95	23:00h	100.0/100	100%
11070000	)-0000-4000-8000-0000C0000R01	PS51,243.16	12:00h	77.3/100	100%
14230000	)-0000-4000-8000-0000C0000R01	PS58,515.38	19:00h	50.1/100	100%
13430000	)-0000-4000-8000-0000C0000R01	PS69,754.11	19:00h	58.1/100	100%
11270000	)-0000-4000-8000-0000C0000R01	PS70,676.12	23:00h	69.8/100	100%
13330000	)-0000-4000-8000-0000C0000R01	PS83,829.78	1:00h	91.7/100	100%
12930000	)-0000-4000-8000-0000C0000R01	PS104,030.83	3:00h	100.0/100	100%
10760000	)-0000-4000-8000-0000C0000R01	PS92,817.17	5:00h	97.9/100	100%
13350000	)-0000-4000-8000-0000C0000R01	PS81,493.28	13:00h	81.2/100	100%
11370000	)-0000-4000-8000-0000C0000R01	PS85,716.20	17:00h	81.3/100	100%
11040000	)-0000-4000-8000-0000C0000R01	PS118,293.94	21:00h	87.6/100	100%
11350000	)-0000-4000-8000-0000C0000R01	PS70,438.53	22:00h	52.5/100	100%
10000000	)-0000-4000-8000-0000C0000R01	PS111,641.07	23:00h	93.6/100	100%
14260000	)-0000-4000-8000-0000C0000R01	PS81,263.94	21:00h	39.8/100	100%
12330000	)-0000-4000-8000-0000C0000R01	PS88,696.12	8:00h	100.0/100	100%
12200000	)-0000-4000-8000-0000C0000R01	PS87,852.70	5:00h	84.8/100	100%
10330000	)-0000-4000-8000-0000C0000R01	PS89,518.91	6:00h	84.9/100	100%
14070000	)-0000-4000-8000-0000C0000R01	PS89,084.90	19:00h	60.2/100	100%
12300000	)-0000-4000-8000-0000C0000R01	PS89,848.91	0:00h	98.3/100	100%
12070000	)-0000-4000-8000-0000C0000R01	PS80,975.20	5:00h	88.5/100	100%
10340000	)-0000-4000-8000-0000C0000R01	PS89,871.56	23:00h	65.6/100	100%
10330000	)-0000-4000-8000-0000C0000R01	PS82,049.55	7:00h	100.0/100	100%
14250000	)-0000-4000-8000-0000C0000R01	PS105,240.35	1:00h	97.1/100	100%
12630000	)-0000-4000-8000-0000C0000R01	PS116,682.03	4:00h	100.0/100	100%

13530000	)-0000-4000-8000-0000C0000	PS23,601.50	7:00h	88.1/100	100%
14020000	)-0000-4000-8000-0000C0000	PS26,831.77	8:00h	100.0/100	100%
10200000	)-0000-4000-8000-0000C0000	PS24,968.73	22:00h	83.9/100	100%
11940000	)-0000-4000-8000-0000C0000	PS22,933.39	1:00h	88.5/100	100%
10060000	)-0000-4000-8000-0000C0000	PS29,829.78	18:00h	96.9/100	100%
10250000	)-0000-4000-8000-0000C0000	PS28,609.02	20:00h	59.7/100	100%
13000000	)-0000-4000-8000-0000C0000	PS23,404.60	9:00h	93.3/100	100%
11000000	)-0000-4000-8000-0000C0000	PS22,670.96	7:00h	82.0/100	100%
14230000	)-0000-4000-8000-0000C0000	PS26,212.80	6:00h	65.3/100	100%
10820000	)-0000-4000-8000-0000C0000	PS29,419.99	6:00h	86.8/100	100%
13470000	)-0000-4000-8000-0000C0000	PS27,863.57	21:00h	46.0/100	100%
10630000	)-0000-4000-8000-0000C0000	PS28,138.36	4:00h	100.0/100	100%
11770000	)-0000-4000-8000-0000C0000	PS21,722.45	20:00h	90.3/100	100%
11530000	)-0000-4000-8000-0000C0000	PS23,624.54	0:00h	70.0/100	100%
13390000	)-0000-4000-8000-0000C0000	PS28,376.13	3:00h	77.8/100	100%
14090000	)-0000-4000-8000-0000C0000	PS29,862.25	1:00h	87.9/100	100%
14360000	)-0000-4000-8000-0000C0000	PS26,211.52	5:00h	67.6/100	100%
10740000	)-0000-4000-8000-0000C0000	PS20,317.24	3:00h	91.4/100	100%
13960000	)-0000-4000-8000-0000C0000	PS28,851.86	9:00h	80.4/100	100%
14470000	)-0000-4000-8000-0000C0000	PS25,252.28	1:00h	70.9/100	100%
12300000	)-0000-4000-8000-0000C0000	PS21,168.55	19:00h	80.3/100	100%
11340000	)-0000-4000-8000-0000C0000	PS22,436.47	3:00h	94.5/100	100%
14030000	)-0000-4000-8000-0000C0000	PS21,278.13	4:00h	100.0/100	100%
12370000	)-0000-4000-8000-0000C0000	PS23,239.32	5:00h	72.1/100	100%
12940000	)-0000-4000-8000-0000C0000	PS25,497.66	3:00h	62.7/100	100%
14330000	)-0000-4000-8000-0000C0000	PS31,166.69	18:00h	95.1/100	100%
10060000	)-0000-4000-8000-0000C0000	PS22,048.00	9:00h	97.6/100	100%
13300000	)-0000-4000-8000-0000C0000	PS29,689.55	22:00h	94.1/100	100%
12460000	)-0000-4000-8000-0000C0000	PS31,664.96	22:00h	68.6/100	100%
13290000	)-0000-4000-8000-0000C0000	PS30,671.96	4:00h	96.9/100	100%
13060000	)-0000-4000-8000-0000C0000	PS30,562.88	18:00h	49.6/100	100%
11030000	)-0000-4000-8000-0000C0000	PS22,433.15	20:00h	41.6/100	100%
12920000	)-0000-4000-8000-0000C0000	PS29,433.31	0:00h	100.0/100	100%
10000000	)-0000-4000-8000-0000C0000	PS20,325.54	17:00h	91.1/100	100%
14900000	)-0000-4000-8000-0000C0000	PS36,611.43	11:00h	78.8/100	100%
12560000	)-0000-4000-8000-0000C0000	PS35,041.89	23:00h	73.8/100	100%
10500000	)-0000-4000-8000-0000C0000	PS22,980.28	1:00h	59.7/100	100%
14060000	)-0000-4000-8000-0000C0000	PS21,760.55	5:00h	100.0/100	100%

13440000	)-0000-4000-8000-0000C0000	PS375,490.52	22:00h	57.5/100	100%
12740000	)-0000-4000-8000-0000C0000	PS34,247.99	18:00h	60.3/100	100%
14580000	)-0000-4000-8000-0000C0000	PS36,579.58	19:00h	66.9/100	100%
12080000	)-0000-4000-8000-0000C0000	PS30,884.02	2:00h	92.4/100	100%
13770000	)-0000-4000-8000-0000C0000	PS35,271.66	5:00h	71.6/100	100%
14680000	)-0000-4000-8000-0000C0000	PS32,651.15	13:00h	96.8/100	100%
13640000	)-0000-4000-8000-0000C0000	PS34,404.29	17:00h	81.5/100	100%
11380000	)-0000-4000-8000-0000C0000	PS30,739.47	2:00h	98.3/100	100%
13460000	)-0000-4000-8000-0000C0000	PS32,979.61	11:00h	51.4/100	100%
11380000	)-0000-4000-8000-0000C0000	PS38,838.05	1:00h	99.2/100	100%
12020000	)-0000-4000-8000-0000C0000	PS33,748.03	9:00h	66.8/100	100%
14610000	)-0000-4000-8000-0000C0000	PS35,280.13	18:00h	100.0/100	100%
11350000	)-0000-4000-8000-0000C0000	PS308,711.13	22:00h	94.1/100	100%
12660000	)-0000-4000-8000-0000C0000	PS39,152.61	1:00h	100.0/100	100%
10690000	)-0000-4000-8000-0000C0000	PS34,205.45	23:00h	68.2/100	100%
11650000	)-0000-4000-8000-0000C0000	PS33,490.25	2:00h	100.0/100	100%
13220000	)-0000-4000-8000-0000C0000	PS301,051.07	0:00h	100.0/100	100%
12980000	)-0000-4000-8000-0000C0000	PS35,189.83	3:00h	68.4/100	100%
13700000	)-0000-4000-8000-0000C0000	PS39,115.69	5:00h	62.2/100	100%
13080000	)-0000-4000-8000-0000C0000	PS34,564.23	2:00h	87.8/100	100%
13580000	)-0000-4000-8000-0000C0000	PS3014,801.64	11:00h	100.0/100	100%
12120000	)-0000-4000-8000-0000C0000	PS36,980.03	23:00h	70.6/100	100%
14340000	)-0000-4000-8000-0000C0000	PS304,923.84	15:00h	99.4/100	100%
11750000	)-0000-4000-8000-0000C0000	PS39,279.25	0:00h	96.0/100	100%
11450000	)-0000-4000-8000-0000C0000	PS3014,880.18	3:00h	100.0/100	100%
12640000	)-0000-4000-8000-0000C0000	PS33,365.67	1:00h	95.7/100	100%
14570000	)-0000-4000-8000-0000C0000	PS35,813.45	2:00h	89.3/100	100%
10090000	)-0000-4000-8000-0000C0000	PS38,382.36	4:00h	89.4/100	100%
14980000	)-0000-4000-8000-0000C0000	PS32,227.63	22:00h	71.2/100	100%
10780000	)-0000-4000-8000-0000C0000	PS37,404.24	0:00h	100.0/100	100%
11570000	)-0000-4000-8000-0000C0000	PS309,247.51	22:00h	100.0/100	100%
11080000	)-0000-4000-8000-0000C0000	PS38,619.66	23:00h	100.0/100	100%
11550000	)-0000-4000-8000-0000C0000	PS32,100.97	21:00h	57.8/100	100%
14820000	)-0000-4000-8000-0000C0000	PS370,684.90	5:00h	68.3/100	100%
12910000	)-0000-4000-8000-0000C0000	PS377,569.06	3:00h	72.6/100	100%
12780000	)-0000-4000-8000-0000C0000	PS348,144.26	7:00h	99.4/100	100%
12220000	)-0000-4000-8000-0000C0000	PS377,362.46	4:00h	82.5/100	100%
11640000	)-0000-4000-8000-0000C0000	PS30118,034.57	3:00h	100.0/100	100%

10090000	)-0000-4000-8000-0000C000R01	11,250.69	2:00h	100.0/100	100%
12580000	)-0000-4000-8000-0000C000R04	45,198.09	0:00h	83.7/100	100%
11820000	)-0000-4000-8000-0000C000R04	43,831.23	1:00h	81.4/100	100%
11040000	)-0000-4000-8000-0000C000R04	49,374.09	20:00h	76.4/100	100%
10590000	)-0000-4000-8000-0000C000R04	45,530.58	9:00h	95.5/100	100%
12440000	)-0000-4000-8000-0000C000R04	41,585.43	0:00h	73.6/100	100%
11920000	)-0000-4000-8000-0000C000R01	47,809.19	2:00h	100.0/100	100%
12700000	)-0000-4000-8000-0000C000R04	44,586.98	7:00h	63.4/100	100%
10020000	)-0000-4000-8000-0000C000R04	41,341.67	20:00h	68.1/100	100%
10790000	)-0000-4000-8000-0000C000R04	44,573.08	4:00h	100.0/100	100%
14400000	)-0000-4000-8000-0000C000R04	40,623.70	5:00h	69.3/100	100%
13550000	)-0000-4000-8000-0000C000R01	48,549.42	22:00h	100.0/100	100%
14080000	)-0000-4000-8000-0000C000R04	48,054.27	23:00h	84.1/100	100%
11900000	)-0000-4000-8000-0000C000R04	40,582.90	1:00h	79.8/100	100%
14820000	)-0000-4000-8000-0000C000R04	41,694.25	1:00h	73.7/100	100%
14270000	)-0000-4000-8000-0000C000R04	44,089.60	2:00h	67.2/100	100%
12950000	)-0000-4000-8000-0000C000R04	48,235.47	23:00h	77.5/100	100%
14570000	)-0000-4000-8000-0000C000R04	45,684.17	19:00h	79.3/100	100%
11070000	)-0000-4000-8000-0000C000R04	46,291.89	0:00h	98.5/100	100%
10580000	)-0000-4000-8000-0000C000R04	44,347.58	3:00h	59.9/100	100%
10050000	)-0000-4000-8000-0000C000R01	46,526.99	8:00h	92.4/100	100%
10840000	)-0000-4000-8000-0000C000R01	41,218.89	13:00h	91.3/100	100%
13580000	)-0000-4000-8000-0000C000R01	43,606.82	0:00h	100.0/100	100%
13550000	)-0000-4000-8000-0000C000R04	46,675.07	4:00h	88.0/100	100%
11850000	)-0000-4000-8000-0000C000R04	48,742.86	14:00h	60.1/100	100%
12590000	)-0000-4000-8000-0000C000R04	49,426.78	23:00h	78.6/100	100%
12390000	)-0000-4000-8000-0000C000R01	44,969.11	23:00h	100.0/100	100%
10580000	)-0000-4000-8000-0000C000R04	43,886.44	21:00h	72.0/100	100%
13940000	)-0000-4000-8000-0000C000R01	45,251.87	3:00h	100.0/100	100%
14440000	)-0000-4000-8000-0000C000R04	44,410.10	3:00h	74.7/100	100%
10770000	)-0000-4000-8000-0000C000R04	42,412.30	20:00h	69.6/100	100%
10940000	)-0000-4000-8000-0000C000R04	40,841.14	8:00h	53.3/100	100%
13500000	)-0000-4000-8000-0000C000R04	43,018.00	9:00h	94.3/100	100%
12850000	)-0000-4000-8000-0000C000R01	46,768.36	6:00h	100.0/100	100%
11550000	)-0000-4000-8000-0000C000R04	48,158.55	6:00h	68.1/100	100%
10500000	)-0000-4000-8000-0000C000R04	44,488.57	3:00h	68.3/100	100%
14800000	)-0000-4000-8000-0000C000R04	41,445.11	4:00h	92.5/100	100%
12250000	)-0000-4000-8000-0000C000R04	48,656.26	20:00h	68.8/100	100%

11090000	)-0000-4000-8000-0000C0000	RS40,352.61	21:00h	76.2/100	100%
11000000	)-0000-4000-8000-0000C0000	RS43,292.96	0:00h	77.3/100	100%
10980000	)-0000-4000-8000-0000C0000	RS45,732.57	0:00h	100.0/100	100%
13240000	)-0000-4000-8000-0000C0000	RS48,206.42	5:00h	83.4/100	100%
10460000	)-0000-4000-8000-0000C0000	RS40,065.74	2:00h	87.5/100	100%
12060000	)-0000-4000-8000-0000C0000	RS41,921.84	3:00h	92.5/100	100%
13380000	)-0000-4000-8000-0000C0000	RS45,458.04	20:00h	75.0/100	100%
11250000	)-0000-4000-8000-0000C0000	RS42,087.47	22:00h	96.8/100	100%
12050000	)-0000-4000-8000-0000C0000	RS42,466.60	4:00h	100.0/100	100%
58000000	)-0000-4000-8000-0000C0000	RS428.95	15:00h	9.1/100	0%
12280000	)-0000-4000-8000-0000C0000	RS59,055.64	7:00h	84.0/100	0%
11200000	)-0000-4000-8000-0000C0000	RS51,328.31	13:00h	55.1/100	100%
13380000	)-0000-4000-8000-0000C0000	RS54,099.65	5:00h	57.5/100	100%
11780000	)-0000-4000-8000-0000C0000	RS57,555.32	7:00h	83.8/100	100%
12010000	)-0000-4000-8000-0000C0000	RS51,265.48	7:00h	87.5/100	100%
10420000	)-0000-4000-8000-0000C0000	RS72,939.83	2:00h	73.4/100	100%
10810000	)-0000-4000-8000-0000C0000	RS70,326.53	16:00h	84.7/100	100%
13260000	)-0000-4000-8000-0000C0000	RS72,176.29	2:00h	72.5/100	100%
13360000	)-0000-4000-8000-0000C0000	RS714,386.78	20:00h	86.1/100	100%
14000000	)-0000-4000-8000-0000C0000	RS55,317.67	1:00h	90.3/100	100%
12490000	)-0000-4000-8000-0000C0000	RS51,638.66	4:00h	90.9/100	100%
11680000	)-0000-4000-8000-0000C0000	RS59,503.84	2:00h	89.8/100	100%
12620000	)-0000-4000-8000-0000C0000	RS52,917.72	23:00h	60.3/100	100%
14380000	)-0000-4000-8000-0000C0000	RS70,564.10	3:00h	71.2/100	100%
10800000	)-0000-4000-8000-0000C0000	RS55,284.81	8:00h	77.1/100	100%
12400000	)-0000-4000-8000-0000C0000	RS58,803.03	20:00h	75.8/100	100%
12080000	)-0000-4000-8000-0000C0000	RS55,918.00	22:00h	76.5/100	100%
12460000	)-0000-4000-8000-0000C0000	RS713,105.10	23:00h	100.0/100	100%
12570000	)-0000-4000-8000-0000C0000	RS78,994.37	20:00h	62.2/100	100%
12470000	)-0000-4000-8000-0000C0000	RS58,329.62	13:00h	87.0/100	100%
12840000	)-0000-4000-8000-0000C0000	RS76,684.45	5:00h	100.0/100	100%
14080000	)-0000-4000-8000-0000C0000	RS53,147.75	11:00h	73.3/100	100%
13200000	)-0000-4000-8000-0000C0000	RS70,652.63	7:00h	94.1/100	100%
14040000	)-0000-4000-8000-0000C0000	RS57,460.98	19:00h	82.2/100	100%
10280000	)-0000-4000-8000-0000C0000	RS53,972.61	9:00h	95.9/100	100%
13250000	)-0000-4000-8000-0000C0000	RS57,133.60	23:00h	65.3/100	100%
14500000	)-0000-4000-8000-0000C0000	RS712,448.05	0:00h	100.0/100	100%
10810000	)-0000-4000-8000-0000C0000	RS70,277.98	23:00h	92.8/100	100%

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** múltiplas falhas de login antes da transação (força bruta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000002 às 1:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000002 às 21:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** múltiplas falhas de login antes da transação (força bruta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000003 às 18:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000003 às 22:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsevrlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000003 às 20:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000004 às 5:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsevrlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** múltiplas falhas de login antes da transação (força bruta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000004 às 7:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000004 às 8:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

Técnica associada: Brute Force (ID: **T1110**), tática **Credential Access**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000005 às 15:00h.
- **Procedimentos sugeridos:** During Operation Dream Job, Lazarus Group performed brute force attacks against administrator accounts.(Citation: E... | During the 2016 Ukraine Electric Power Attack, Sandworm Team used a script to attempt RPC authentication against a ... | APT38 has used brute force techniques to attempt account access when passwords are unknown or when password hashes are unavailable.(Citation: CISA AA20-239A Be...

Técnica associada: Account Manipulation (ID: **T1098**), tática **Persistence**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** dispositivo novo vinculado + alteração de limite Pix (tomada de conta).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000005 às 7:00h.
- **Procedimentos sugeridos:** During the 2016 Ukraine Electric Power Attack, Sandworm Team used the `sp\_addlinkedsevrlogin` command in MS-SQL to c... | HAFNIUM has granted privileges to domain accounts and reset the password for default admin accounts....

Técnica associada: Data Manipulation (ID: **T1565**), tática **Impact**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** tipo de transação (fallback, sem correlação de log disponível).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000005 às 1:00h.
- **Procedimentos sugeridos:** FIN13 has injected fraudulent transactions into compromised networks that mimic legitimate behavior to siphon off incremental amounts of money.(Citation: Sygni... | PHASEJAM has blocked legitimate upgrades of Ivanti Connect Secure systems and falsely indicates a successful upgrade while operating on an older version.(Cit...

Técnica associada: Valid Accounts (ID: **T1078**), tática **Stealth**. Fonte: banco de dados (dinâmico).

- **Critério de correlação:** mudança de localização entre acessos recentes (uso de credencial fora do padrão).
- **Cliente (pseudônimo):** 10000000-0000-4000-8000-000000000005 às 3:00h.
- **Procedimentos sugeridos:** During RedPenguin, UNC3886 used legitimate credentials to gain privileged access to Juniper routers.(Citation: Mand... | During Operation MidnightEclipse, threat actors extracted sensitive credentials while moving laterally through compromised networks.(Citation: Volexity UPST... | During the 2015 Ukraine Electric Power Attack, Sandworm Team used valid accounts on the corporate network to escala...

4. Validação dos Modelos (transparência metodológica)

Amostra: 1125 transações de treino / 375 de teste. Classificador de triagem: precision=1.00, recall=0.98 (reproduz decisões históricas, não detecta ataques inéditos). Detector selecionado (elliptic envelope): precision=1.00, recall=0.45, F1=0.62 (rótulo usado somente para auditoria comparativa). Regressão de severidade: R^2 (validação cruzada 5-fold)=0.65.

5. Comparação dos Detectores de Anomalia

Modelo	Alertas	Precision	Recall	F1	Tempo (s)
isolation forest	225	1.00	0.45	0.62	0.234
elliptic envelope	225	1.00	0.45	0.62	0.096
one class svm	224	0.88	0.39	0.54	0.040
local outlier factor	208	0.59	0.24	0.34	0.047

O detector selecionado representa o melhor desempenho obtido entre os modelos avaliados neste conjunto de dados, considerando F1-score, recall, precision e tempo de execução. Como a avaliação foi realizada sobre uma base sintética, os resultados demonstram a viabilidade da abordagem proposta e podem servir como referência para estudos futuros utilizando dados reais.